

SmartCampusBuddy

Project Report

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Course / Module: Programming in Java

Introduction

SmartCampusBuddy is a lightweight Flask web application that helps students store study resources, manage schedules, and receive keyword-based recommendations.

Problem Statement

Students often find it hard to locate quality study resources for specific course topics and to keep an organized study schedule.

Functional Requirements

Resource management (add/edit/delete/list), Recommendation engine, Schedule creation and management, Dashboard with analytics.

Non-functional Requirements

Performance: Fast for small datasets. Usability: Simple UI. Maintainability: Modular code. Security: Basic input validation and hashed passwords.

Architecture

Browser -> Flask App -> SQLite + Recommender (TF-IDF).

Design & UML

Included use-case and architecture diagrams.

Testing & CI

Unit tests using pytest. GitHub Actions workflow runs tests on push.

Deployment

Dockerfile provided for containerization.

References

Original project guideline: local file at /mnt/data/BuildYourOwnProject.pdf

Use-case Diagram



Architecture Diagram

